

Orpheus Hindi TTS Model — Performance Evaluation Report

Evaluation Framework: Whisper-large ASR
Language: Hindi (hi-IN) | **Total Samples:** 500
Report Date: July 2026

1. Executive Summary

This report presents a comprehensive evaluation of the **Orpheus Hindi TTS model** across three training stages:

Stage	Description
Base Model	Pre-trained Orpheus TTS without Hindi-specific fine-tuning
Stage 1	Fine-tuned on 75+ hours of Hindi Text-speech data for pronunciation accuracy .
Stage 2	Further optimized for Two speakers: Prakhar and Prerna for male and female voices.

All 500 evaluation samples were synthesized by each model stage, then transcribed by **Whisper-large ASR**. The transcriptions were compared against normalized reference texts using **Word Error Rate (WER)** and **Character Error Rate (CER)**.

Key Findings at a Glance

Metric	Base Model	Stage 1	Stage 2	Δ Base→S2
Mean WER	0.9246	0.3514	0.3304	✓ -64.3%
Mean CER	0.8801	0.1913	0.1707	✓ -80.6%
Median WER	0.3333	0.3060	0.2857	✓ -14.3%
Median CER	0.2330	0.1176	0.1111	✓ -52.3%

INSIGHTS: Fine-tuning reduced mean WER by 64.3% and mean CER by 80.6% from the Base Model to Stage 2. The large gap between mean and median in the Base Model (WER: 0.92 mean vs 0.33 median) indicates a long tail of high-error samples — particularly on short texts where ASR hallucinations inflate scores. Stage 1 and Stage 2 fine-tuning dramatically compressed this tail.

Sample-Level Improvement Distribution (Base → Stage 2, WER)

Outcome	Count	Percentage
✓ Improved	291	58.2%
⚠ Same	14	2.8%
✗ Regressed	195	39.0%

2. Stage-over-Stage Improvement Analysis

This section shows the **absolute and relative improvement** at each transition.

Base Model → Stage 1

Metric	Base	Stage 1	Absolute Δ	Relative Δ
Mean WER	0.9246	0.3514	-0.5732	-62.0%
Mean CER	0.8801	0.1913	-0.6888	-78.3%

- 275 samples improved (55.0%), 211 regressed (42.2%), 14 same (2.8%)

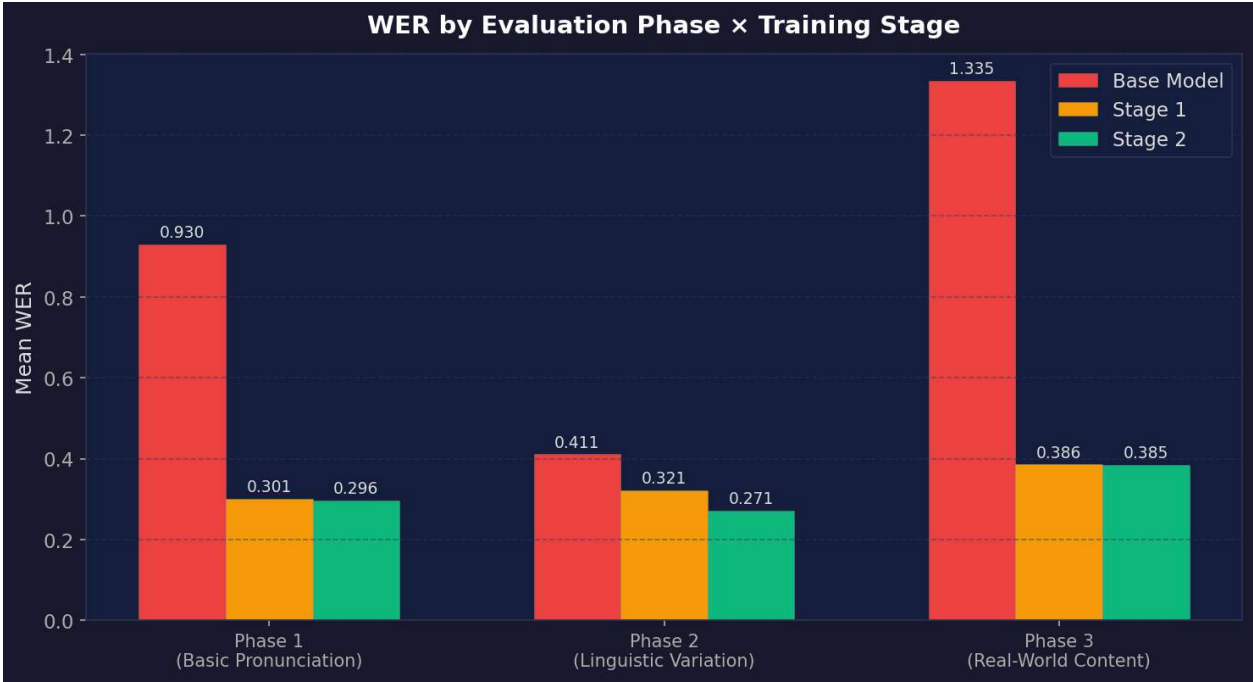
Stage 1 → Stage 2

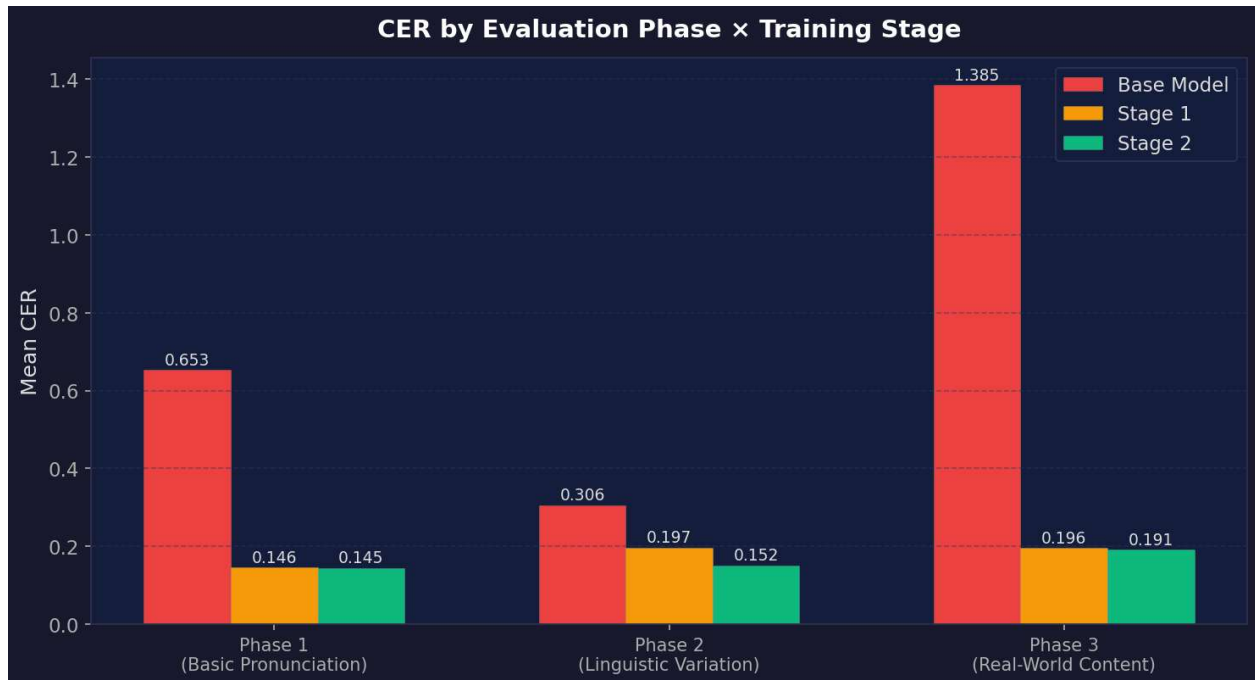
Metric	Stage 1	Stage 2	Absolute Δ	Relative Δ
Mean WER	0.3514	0.3304	-0.0210	-6.0%
Mean CER	0.1913	0.1707	-0.0206	-10.8%

NOTE: Stage 1 provided the largest improvement. Stage 2 (speaker optimization) provided further incremental gains (~6% WER, ~11% CER reduction), which is expected since Stage 2 focuses more on voice quality and naturalness than on pronunciation accuracy.

3. Performance by Evaluation Phase

Phase	N	WER Base	WE R S1	WE R S2	Δ WER (B→S2)	CER Base	CE R S1	CE R S2	Δ CER (B→S2)
Phase 1 — Basic Pronunciation	50	0.9295	0.3013	0.2963	✓ −68.1%	0.6532	0.1464	0.1446	✓ −77.9%
Phase 2 — Linguistic Variation	200	0.4107	0.3212	0.2705	✓ −34.1%	0.3059	0.1970	0.1517	✓ −50.4%
Phase 3 — Real-World Content	250	1.3347	0.3856	0.3851	✓ −71.1%	1.3849	0.1957	0.1912	✓ −86.2%



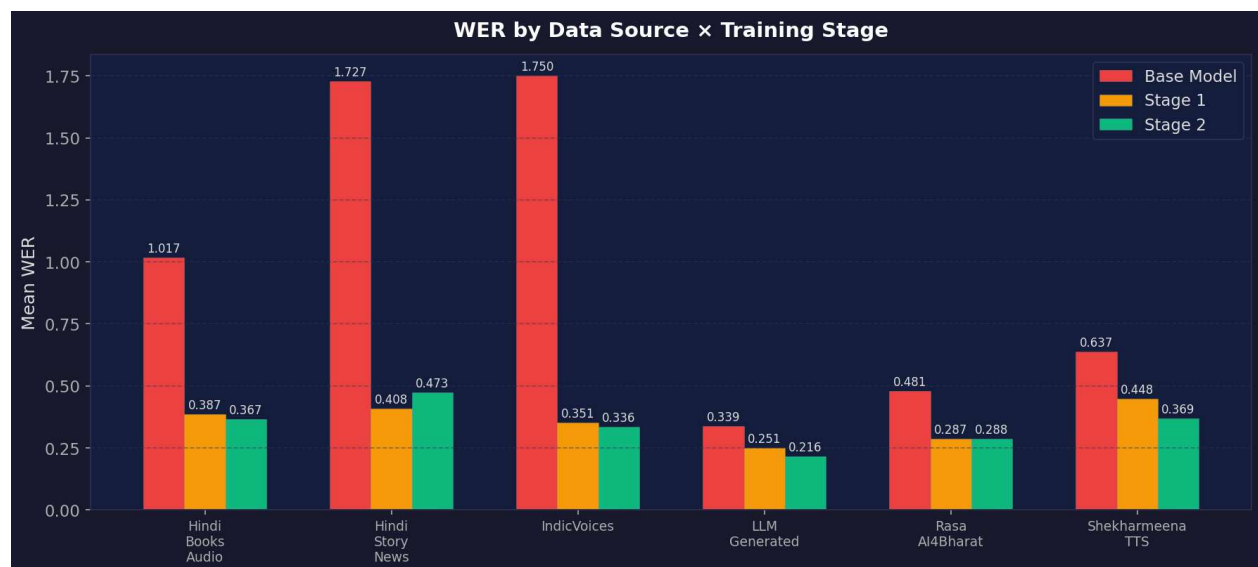


Observations: - Phase 3 (real-world content) had the highest Base Model error (WER 1.33, CER 1.38), but showed the **most dramatic improvement** after fine-tuning (−71% WER, −86% CER). - Phase 2 (linguistic variation) had the lowest Stage 2 WER (0.2705), showing that the fine-tuned model handles varied linguistic constructions well. - Phase 1 (basic pronunciation) saw the lowest Stage 2 errors across all phases, confirming solid foundational pronunciation ability.

4. Performance by Data Source (**source_from**)

Source	N	WER Base	WER S1	WER S2	Δ WER (B→S2)	CER Base	CER S1	CER S2	Δ CER (B→S2)
Hindi_Books_Audio	50	1.0167	0.3872	0.3671	✓ −63.9%	0.9357	0.1740	0.1741	✓ −81.4%

Hindi_Story_News	50	1.7274	0.4082	0.4727	✓ -72.6%	1.5880	0.2039	0.2448	✓ -84.6%
IndicVoices	100	1.7496	0.3513	0.3363	✓ -80.8%	1.9851	0.2122	0.2057	✓ -89.6%
LLM_Generated	80	0.3392	0.2505	0.2162	✓ -36.2%	0.2626	0.1631	0.1288	✓ -51.0%
Rasa_AI4Bharat	110	0.4807	0.2866	0.2876	✓ -40.2%	0.3952	0.1173	0.1106	✓ -72.0%
Shekharmeena_TTS	110	0.6373	0.4478	0.3694	✓ -42.0%	0.4626	0.2690	0.1945	✓ -57.9%

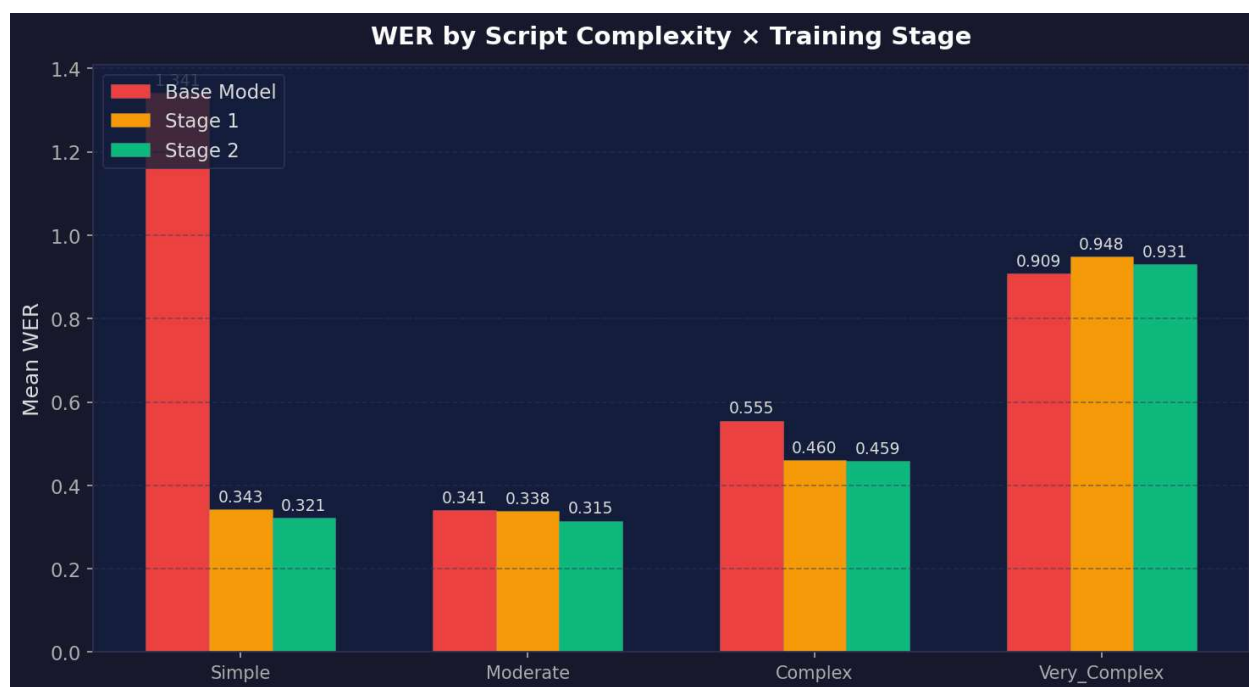


Observations: - **IndicVoices** showed the highest Base Model errors but also the **greatest improvement** (−80.8% WER, −89.6% CER), suggesting the model learned to

generalize from this diverse source well. - **LLM_Generated** samples had the lowest errors across all stages, which is expected since LLM-generated text tends to be cleaner and more structurally consistent. - **Hindi_Story_News** has a notable pattern: Stage 2 WER (0.4727) is higher than Stage 1 (0.4082), indicating **partial regression** in Stage 2 for news/story content — a potential area for targeted improvement. - **Shekharmeena_TTS** consistently has the highest Stage 1 and Stage 2 WER among all sources, suggesting this source's voice characteristics present an ongoing challenge.

5. Performance by Script Complexity

Complexity	N	WER Base	WER S1	WER S2	Δ WER (B→S2)	CER Base	CER S1	CER S2	Δ CER (B→S2)
Simple	284	1.3413	0.3433	0.3213	✓ -76.0%	1.3438	0.1808	0.1662	✓ -87.6%
Moderate	185	0.3410	0.3378	0.3151	✓ -7.6%	0.2413	0.1883	0.1573	✓ -34.8%
Complex	28	0.5552	0.4603	0.4590	✓ -17.3%	0.3969	0.2439	0.2288	✓ -42.4%
Very_Complex	3	0.9087	0.9480	0.9306	⚠ -2.4%	0.8863	0.8863	0.8876	✗ +0.1%



Observations: - **Simple scripts** had the highest Base Model error (inflated by short-text hallucinations) but showed the **most dramatic improvement** (−76% WER). - **Very Complex scripts** (N=3) remain a critical struggle zone. Fine-tuning barely improved them (WER ~0.93 across all stages), and CER actually regressed slightly. These samples involve extremely long, literary Hindi with dense conjuncts and rare vocabulary. - **Complex scripts** (N=28) see moderate improvement (−17% WER), but Stage 1 → Stage 2 improvement is negligible, suggesting speaker optimization doesn't help with complex phonetic challenges.

[!WARNING] **Struggle Zone:** Very_Complex and Complex scripts show minimal improvement from fine-tuning. The model needs targeted training on complex Hindi orthographic patterns (conjuncts, rare characters, literary vocabulary).

6. Performance by Sentence Length Category

Category	N	WER Base	WE R S1	WE R S2	Δ WER (B→S2)	CER Base	CE R S1	CE R S2	Δ CER (B→S2)
Short (1–5 words)	70	3.9528	0.4117	0.4617	✓ –88.3%	4.3220	0.1606	0.2234	✓ –94.8%
Medium (6–10 words)	183	0.5470	0.3375	0.3008	✓ –45.0%	0.4202	0.1859	0.1530	✓ –63.6%
Long (11–20 words)	129	0.3610	0.3026	0.2887	✓ –20.0%	0.2501	0.1510	0.1268	✓ –49.3%
Very_Long (21+ words)	118	0.3299	0.3908	0.3440	✗ +4.3%	0.2404	0.2619	0.2151	✓ –10.5%

Observations: - **Short texts** had extremely high Base Model WER (3.95) due to ASR hallucinations — the model would generate extra words for very short inputs. Fine-tuning reduced this dramatically (–88%), but Stage 2 WER (0.46) is still the highest among all categories. - **Medium and Long texts** show the best Stage 2 performance (WER 0.30 and 0.29 respectively), representing the "sweet spot" for the model. - **Very_Long texts** show a **regression pattern**: Stage 1 WER (0.39) and Stage 2 WER (0.34) are both *higher* than Base Model WER (0.33). The fine-tuning seems to have introduced some degradation for very long sequences.

[!CAUTION] **Regression on Very Long Texts**: WER increased by +4.3% for Very_Long sentences (Base→S2). This suggests the fine-tuning data may not have contained sufficient long-form content, causing the model to lose some of its original long-text generation capability.

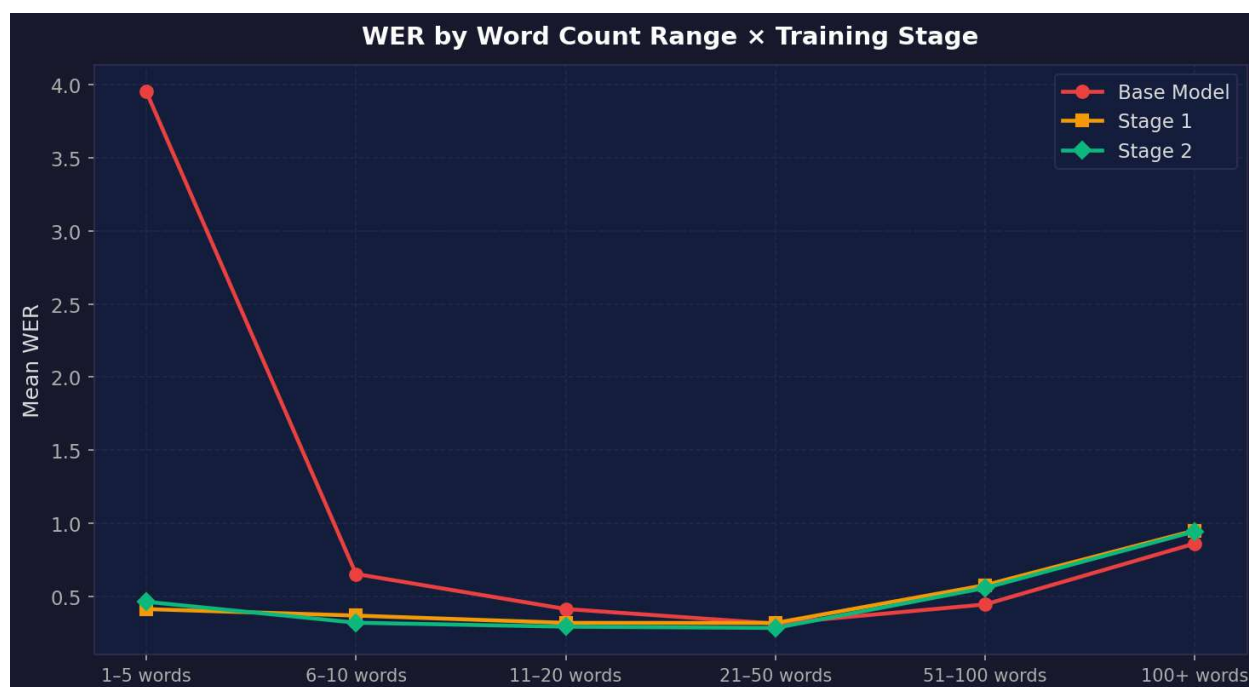
7. Performance by Code Mixing Ratio

Bucket	N	WER Base	WE R S1	WE R S2	Δ WER (B→S2)	CER Base	CE R S1	CE R S2	Δ CER (B→S2)
0.00 (None)	394	1.0040	0.3346	0.3252	✓ -67.6%	0.9885	0.1659	0.1552	✓ -84.3%
0.01–0.10 (Low)	30	0.4127	0.4379	0.2961	✓ -28.2%	0.3417	0.3202	0.1965	✓ -42.5%
0.11–0.30 (Moderate)	51	0.4731	0.4079	0.3353	✓ -29.1%	0.3922	0.2824	0.2155	✓ -45.0%
0.31–0.50 (High)	22	1.1834	0.4495	0.3707	✓ -68.7%	0.7973	0.2847	0.2265	✓ -71.6%
0.50+ (Very High)	3	1.3935	0.0238	0.9788	✓ -29.8%	0.9306	0.0105	0.7878	✓ -15.3%

Observations: - **No code mixing** (pure Hindi) accounts for 78.8% of samples and shows strong improvement (-67.6% WER). - **High code mixing** (0.31–0.50) shows a dramatic improvement (-68.7% WER), comparable to pure Hindi, suggesting the model handles Hindi-English mixing well after fine-tuning. - **Very High code mixing** (0.50+, N=3) is interesting: Stage 1 achieved near-perfect WER (0.024) but Stage 2 regressed significantly (0.979). This may be due to the small sample size or speaker optimization interfering with English-dominant synthesis. - **Low and Moderate mixing** show smaller but consistent improvements, with Stage 2 performing better than Stage 1 in all cases.

8. Performance by Word Count (Interval Buckets)

Word Range	N	WER Base	WER S1	WER S2	Δ WER (B→S2)	CER Base	CER S1	CER S2	Δ CER (B→S2)
1–5	70	3.9528	0.4117	0.4617	✓ –88.3%	4.3220	0.1606	0.2234	✓ –94.8%
6–10	83	0.6520	0.3673	0.3184	✓ –51.2%	0.5118	0.1955	0.1465	✓ –71.4%
11–20	189	0.4123	0.3173	0.2912	✓ –29.4%	0.2984	0.1693	0.1457	✓ –51.2%
21–50	139	0.3126	0.3166	0.2826	✓ –9.6%	0.2144	0.1832	0.1402	✓ –34.6%
51–100	15	0.4437	0.5756	0.5564	✗ +25.4%	0.3648	0.4790	0.4683	✗ +28.4%
100+	4	0.8604	0.9498	0.9432	✗ +9.6%	0.8406	0.8851	0.8811	✗ +4.8%



Observations: - The model performs best on texts with **6–50 words** (WER 0.28–0.32 at Stage 2). - **Very short texts (1–5 words)** show the most improvement from Base (–88%) but still have the highest Stage 2 WER among the primary ranges due to hallucination artifacts. - **51–100 word texts** are a clear **regression zone**: Stage 2 WER (0.5564) is 25.4% *worse* than Base Model (0.4437). This is the most concerning finding. - **100+ word texts** also regress, though with only 4 samples this finding is less statistically significant.

[!WARNING] **Critical Regression:** Texts with 51–100 words show a 25.4% WER increase after fine-tuning. This represents a significant quality degradation for medium-long content and should be prioritized for investigation.

9. Performance by Character Count (Interval Buckets)

Char Range	N	WER Base	WER S1	WER S2	Δ WER (B $\rightarrow$ S2)	CER Base	CER S1	CER S2	Δ CER (B $\rightarrow$ S2)
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1–20	4 8	4.7386	0.40 31	0.47 08	✓ –90.1%	5.473 2	0.15 20	0.26 46	✓ –95.2%
21–50	1 0 9	0.9692	0.36 51	0.33 69	✓ –65.2%	0.775 1	0.18 20	0.14 34	✓ –81.5%
51–100	1 8 4	0.4143	0.31 50	0.28 69	✓ –30.8%	0.299 9	0.16 99	0.14 22	✓ –52.6%
101–200	1 2 5	0.3060	0.30 46	0.27 71	✓ –9.4%	0.205 9	0.16 68	0.12 96	✓ –37.1%
200–500	3 0	0.3761	0.55 83	0.48 89	✗ +30.0%	0.285 7	0.42 88	0.37 22	✗ +30.2%
500+	4	0.8604	0.94 98	0.94 32	✗ +9.6%	0.840 6	0.88 51	0.88 11	✗ +4.8%

Observations: - The model's **optimal range** is 51–200 characters (WER 0.28–0.29 at Stage 2), covering the majority of samples (309 out of 500). - **Very short texts (1–20 chars)** show massive Base Model inflation (WER 4.74) but are well-corrected by fine-tuning (–90%), though Stage 2 slightly regresses from Stage 1. - **200–500 character texts** show a concerning **30% WER regression** from Base to Stage 2. This is consistent with the word count findings and confirms a long-text degradation problem.

10. Key Findings & Discussion

Where the Model Excels ✓

1. **Medium-length Hindi text** (6–50 words / 51–200 chars): Consistent WER < 0.30, CER < 0.15 at Stage 2.
2. **Pure Hindi without code mixing**: Strong performance with 67.6% WER improvement.
3. **LLM-Generated and Rasa_AI4Bharat content**: Lowest error rates, suggesting clean training text translates to better synthesis.
4. **Simple and Moderate script complexity**: Good handling of standard Devanagari with consistent improvement.

Where the Model Struggles ❌

1. **Very Complex scripts** (N=3): WER ~0.93 across all stages — near-complete failure. Literary Hindi with dense conjuncts and rare vocabulary remains unsolvable.
2. **Long texts (51+ words / 200+ chars)**: Regression of 25–30% in WER after fine-tuning. The model loses coherence on extended passages.
3. **Very short texts (1–5 words)**: While improved from Base, Stage 2 WER (0.46) remains high due to hallucinated content.
4. **Hindi_Story_News source**: Stage 2 regresses from Stage 1, suggesting news/literary content doesn't benefit from speaker optimization.
5. **Shekharmeena_TTS source**: Consistently highest error rates across Stage 1 and Stage 2 among all sources.

Recommendations for Future Training

1. **Increase long-text representation**: Add more 50–200 word samples to fine-tuning data to prevent regression.
2. **Target Very Complex scripts**: Curate training data with literary Hindi, dense conjuncts, and rare grapheme clusters.
3. **Address short-text hallucinations**: Consider architectural or decoding strategy changes (e.g., length-constrained generation) for very short inputs.
4. **Source-specific fine-tuning**: Investigate why Shekharmeena_TTS and Hindi_Story_News show higher error rates — potentially domain-specific vocabulary gaps.

11. Sample Spotlights

Best Improvement: HI00334

Metric	Base	Stage 1	Stage 2
WER	27.3333	0.0000	0.0000

- **Phase:** Phase 3 (Real-World Content) | **Source:** IndicVoices
- The Base Model hallucinated extensively (WER 27.33x), but fine-tuning completely resolved it.

Worst Regression: HI00373

Metric	Base	Stage 1	Stage 2
WER	0.2973	0.5263	1.6316

- **Phase:** Phase 3 (Real-World Content) | **Source:** Shekharmeena_TTS
- A well-performing Base Model sample that degraded significantly after fine-tuning. Likely overfitting or domain mismatch in the fine-tuning data.

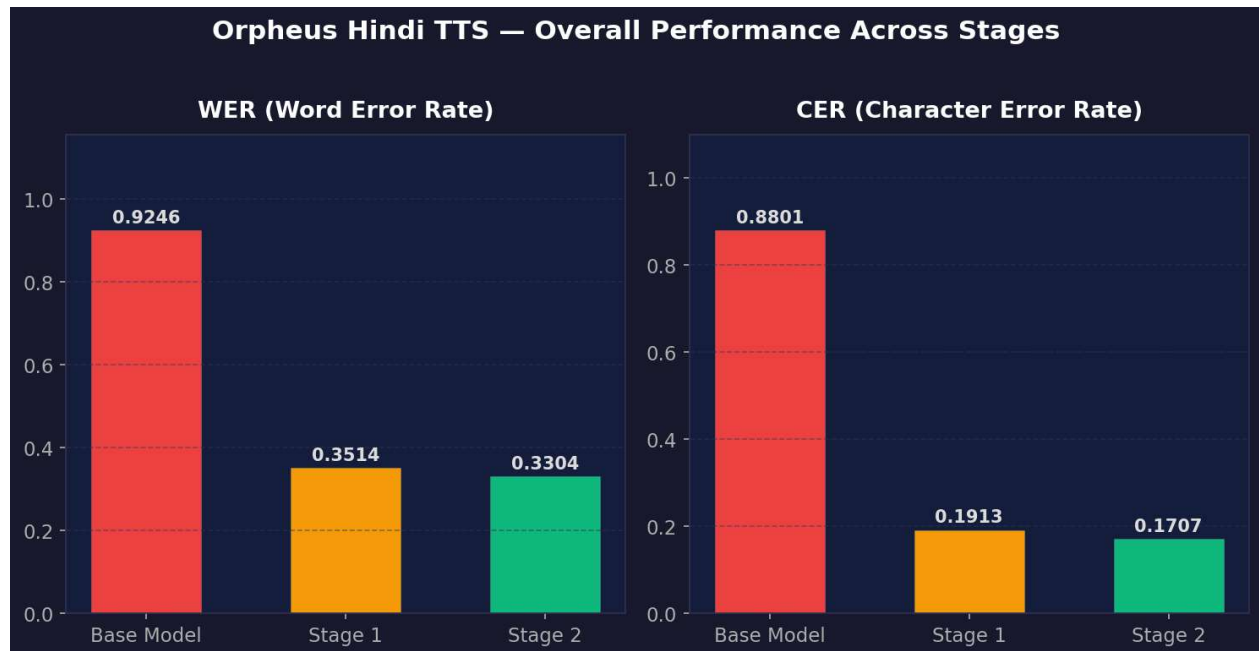
Hardest Sample (Stage 2): HI00042

Metric	Stage 2
WER	2.0000
CER	1.9000

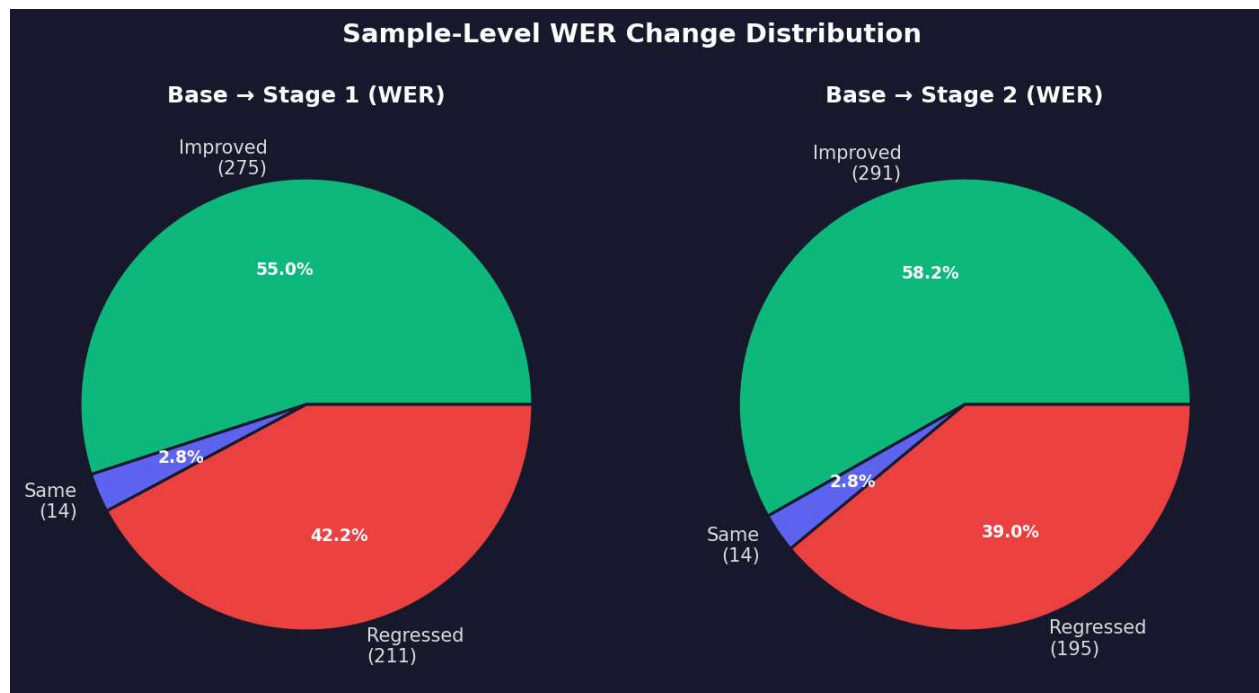
- The model generates significantly more content than the reference, indicating a persistent hallucination issue on this particular sample.
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12. Visualizations

Overall Performance



Sample-Level Improvement Distribution



13. Full Per-Sample Reference Table

#	Sample ID	Phase	Source	WER Base	CER Base	WER S1	CER S1	WER S2	CER S2
1	HI00001	P1	Rasa_AI4B harat	0.0000	0.0000	1.0000	0.2500	0.0000	0.0000
2	HI00002	P1	Rasa_AI4B harat	0.3889	0.1818	0.2778	0.0758	0.1667	0.0758
3	HI00003	P1	Rasa_AI4B harat	2.7143	2.7206	0.2857	0.1176	0.2143	0.0882
4	HI00004	P1	Rasa_AI4B harat	0.1714	0.1034	0.1714	0.1034	0.1714	0.0690
5	HI00005	P1	Rasa_AI4B harat	0.2000	0.0824	0.2000	0.0471	0.2000	0.0471
6	HI00006	P1	Rasa_AI4B harat	0.3000	0.1613	0.3500	0.1452	0.3500	0.1452

7	HI00007	P 1	Rasa_AI4B harat	0.1500	0.0588	0.100 0	0.014 7	0.050 0	0.0441
8	HI00008	P 1	Rasa_AI4B harat	1.0000	0.9200	0.000 0	0.000 0	0.000 0	0.0000
9	HI00009	P 1	Rasa_AI4B harat	0.2174	0.0857	0.304 3	0.100 0	0.260 9	0.0857
1 0	HI00010	P 1	Rasa_AI4B harat	0.5385	0.3846	0.230 8	0.153 8	0.307 7	0.1923
1 1	HI00011	P 1	Rasa_AI4B harat	1.0000	0.7500	0.000 0	0.000 0	0.000 0	0.0000
1 2	HI00012	P 1	Rasa_AI4B harat	0.4375	0.3200	0.187 5	0.080 0	0.312 5	0.0800
1 3	HI00013	P 1	Rasa_AI4B harat	0.2593	0.1429	0.296 3	0.131 0	0.222 2	0.0952
1 4	HI00014	P 1	Rasa_AI4B harat	0.5500	0.2373	0.250 0	0.084 7	0.200 0	0.1017
1 5	HI00015	P 1	Rasa_AI4B harat	0.3333	0.1220	0.333 3	0.122 0	0.166 7	0.0244
1 6	HI00016	P 1	Rasa_AI4B harat	0.1111	0.0541	0.185 2	0.054 1	0.148 1	0.0541

1 7	HI00017	P 1	Rasa_AI4B harat	1.0000	0.5000	0.000 0	0.000 0	0.000 0	0.0000
1 8	HI00018	P 1	Rasa_AI4B harat	0.5417	0.3585	0.375 0	0.169 8	0.375 0	0.1698
1 9	HI00019	P 1	Rasa_AI4B harat	5.7500	6.3333	0.500 0	0.000 0	0.000 0	0.0000
2 0	HI00020	P 1	Rasa_AI4B harat	0.2692	0.0714	0.384 6	0.071 4	0.384 6	0.0714

500 samples with scores across all three stages. Sorted by Sample ID.[!NOTE] The first 20 samples are shown above for quick reference. The complete 500-sample dataset is available in the companion data file at analysis_stats.json under the `per_sample` key.

Appendix: Methodology

Evaluation Pipeline

1. **Text Input** → Orpheus TTS (Base / Stage 1 / Stage 2) → **Audio Output**
2. **Audio Output** → Whisper-large ASR → **Transcription**
3. **Transcription** vs. **Normalized Reference Text** → WER and CER computation

Metrics Definition

- **WER (Word Error Rate):** $(\text{Substitutions} + \text{Insertions} + \text{Deletions}) / \text{Reference Word Count}$
- **CER (Character Error Rate):** $(\text{Substitutions} + \text{Insertions} + \text{Deletions}) / \text{Reference Character Count}$

Normalization

- Text was normalized by removing punctuation, standardizing nukta/chandrabindu variants, and lowercasing where applicable.
- Both reference and ASR transcription underwent identical normalization before comparison.

Report generated on July 3, 2026. Data source: 500 Hindi evaluation samples across 3 phases.